

MGT 151 FINAL PROJECT - GROUP 2

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Methodology

1. How should ER Games ensure they are in line with the trends of the industry?
 - a. **Classification (Fit Model)** to test whether Critic Scores or User Scores has greater impact on sales
2. Which market besides the US should ER Games target when releasing the game?
 - a. **A/B Testing** to calculate p-value for NA vs EU, NA vs Japan & Japan vs EU
3. What impact does the changing preferences and demographics of the typical consumer have on the focus of ER Games?
 - a. **KPI/Dashboarding** to find Sales per 1000 in all Regions
 - b. **Clustering** to find Cluster with most sales + highest scores

How should ER Games ensure they are in line with the trends of the industry?

▼ Summary of Fit				
RSquare		0.025167		
RSquare Adj		0.015883		
Root Mean Square Error		0.816357		
Mean of Response		0.432165		
Observations (or Sum Wgts)		213		
▼ Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Ratio
Model	2	3.61310	1.80655	2.7107
Error	210	139.95224	0.66644	Prob > F
C. Total	212	143.56533		0.0688
▼ Parameter Estimates				
Term	Estimate	Std Error	t Ratio	Prob> t
Intercept	-0.38296	0.377161	-1.02	0.3111
critic score	0.1203514	0.056224	2.14	0.0335*
user score	-0.009499	0.039759	-0.24	0.8114

1. We used Fit Model Testing using NA Sales and Critic Score
2. Critic Score had P-value of 3.35%
3. User Score had P-value of 81.24%
4. As Critic Score P-value is less than 5% - ER Games should use third-party critic scores - in line with industry-trends
5. Risk: Low R-Square - Accuracy & Fit Issues

Which market besides the US should ER Games target when releasing the game?

North America vs EU	North America vs Japan	Japan vs EU
-28669	-116613	87943
-1.73	-6.81	19.66
96%	100%	0%

1. We used A/B testing to gain comparative insights on regional markets for strategy games.
2. Our analysis found that North America was still the most successful market for strategy games, despite what data analysts suggested earlier.
3. Other than North America however, we found Europe to be the second most successful regional market and should be prioritized over Japan when it comes to marketing and distribution.

What impact does the changing preferences and demographics of the typical consumer have on the focus of ER Games?

Count of Games	1
North America Sales per 1000	7.5
European Union Sales per 1000	3.3
Japan Sales per 1000	0
Platform	PC
Critic Score	9.3 (+-0.5)
Rating	Teenagers & Above

1. Conducted KPI Analysis to Calculate Sales per 1000 for Strategy games
2. Hierarchical Clustering using Sales per 1000 in all regions, Platforms, Genre, Rating & Critic Score
3. Cluster 1 is best, but has only 1 games - Risk

What impact does the changing preferences and demographics of the typical consumer have on the focus of ER Games?

Count of Games	1
North America Sales per 1000	125.4
European Union Sales per 1000	58.8
Japan Sales per 1000	29.7
Platform	Weii
Critic Score	7.5 (+-0.5)
Rating	Everyone

1. Clustering the whole dataset using K-means clustering (too many rows)
2. Sports Games with the characteristics on the right perform better as Cluster 1 is best
3. ER Games should change its focus to Sport Games instead of Strategy
4. Risk: only one game in the cluster

Conclusion

1. Critic Scores used to ensure ER Games is in line with trends of the Industry
2. ER Games should release the game in European and North American Markets
3. ER Games is focused on Strategy based games, however, through clustering we found Sports games with the following characteristics:
 - a. Rating: Everyone
 - b. Critic Score: 7.5(+0.5)
 - c. Platform: Wii
4. However, if ER Games specialize only in strategy games, the game should:
 - a. Rating: Teenage and older
 - b. Critic Score: 9.2(+0.5)
 - c. Platform: PC



Thank You